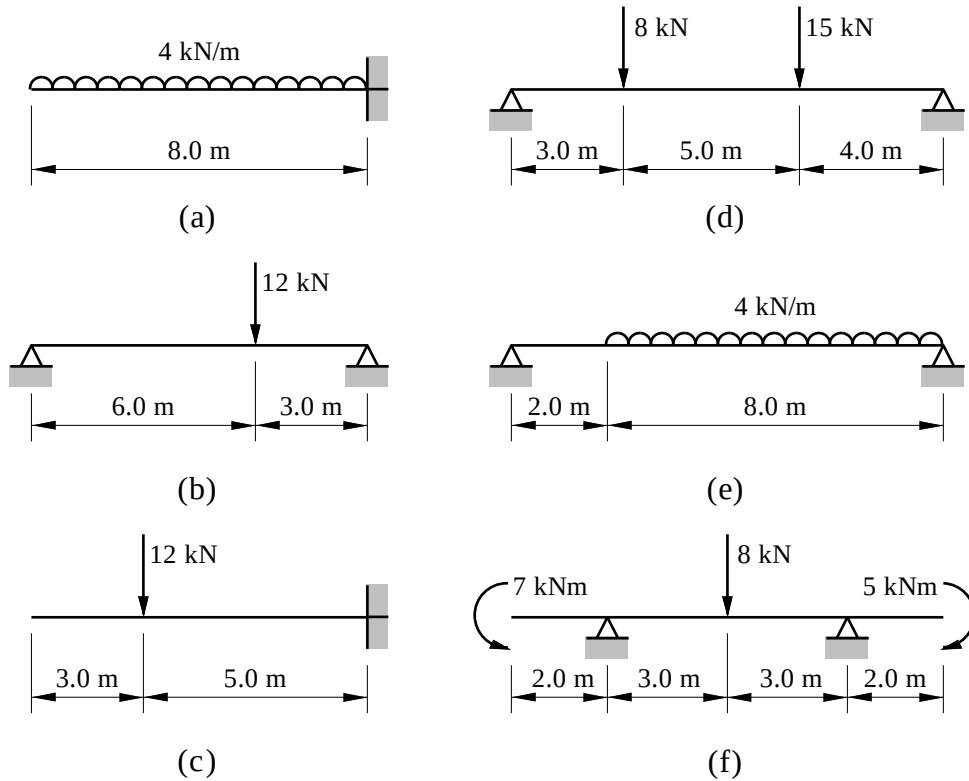


THE UNIVERSITY OF BIRMINGHAM
DEPARTMENT OF CIVIL ENGINEERING

Mechanics 1

Tutorial Sheet - 4

1. Draw the shear force and bending moment diagrams for each of the beams shown below. In each case, find the external reactions, construct the diagrams and mark the position and value of the greatest shear force and the greatest bending moment in the beam.



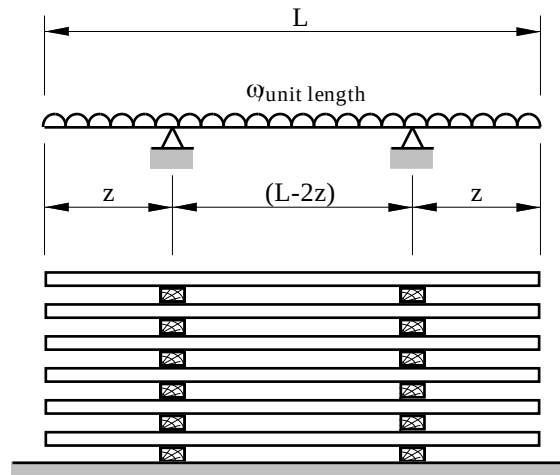
Ans :

	(a)	(b)	(c)	(d)	(e)	(f)
x (m)	8.0	6.0-9.0	3.0-8.0	8.0-12.0	10.0	2.0
F (kN)	-32	-8	-12	-12	-19.2	+4.33
x (m)	8.0	6.0	8.0	8.0	5.2	0.0-2.0
M (kNm)	-128	+24	-60	+48	+46.1	-7.0

2. The lighting standards manufactured by a certain firm consist of a uniform concrete shaft of length L . The finished standards are stacked horizontally in the firm's stockyard, with small wooden packing blocks placed symmetrically between the layers, as shown in the diagram.

In effect, each standard forms a simply-supported beam, carrying its own weight as a uniformly distributed load, ω . Note that the bending moments in each layer will be exactly the same.

Sketch the bending moment diagram for a typical standard, marking peak values in terms of z - the distance of the packing blocks from the ends.

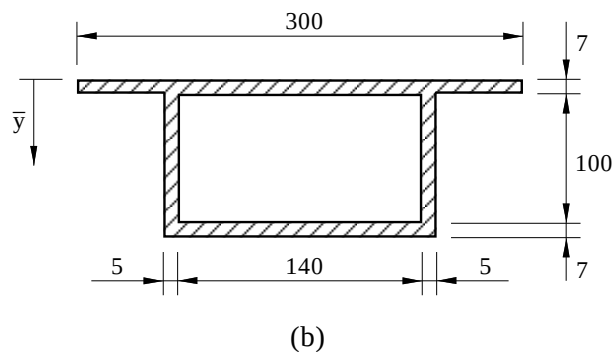
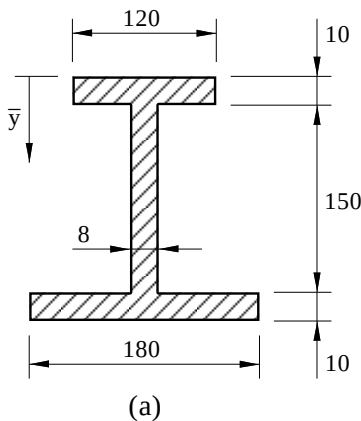


In order to minimise the possibility of damage to the concrete, the *largest* bending moment (positive or negative) is to be as small as possible. At what distance from the ends should the packing blocks be placed?

Hint : consider what happens to the maximum positive bending moment as z is increased - and what happens to the maximum negative bending moment. At what stage is the larger of the two as small as possible?

Ans: $z = 0.207L$

3. For each of the beam sections shown in the diagram (dimensions in millimetres), locate the neutral axis and find the second moment of area of the section.



In each case, find the maximum tensile and compressive stresses in the beam due to a bending moment of +10 kNm.

Ans : (a) $\bar{y} = 96.43 \text{ mm}$, $I = 20.93 \times 10^6 \text{ mm}^4$, $s = +35.2, -46.1 \text{ N/mm}^2$.
 (b) $\bar{y} = 43.46 \text{ mm}$, $I = 9.10 \times 10^6 \text{ mm}^4$, $s = +77.5, -47.8 \text{ N/mm}^2$.

4. The two beam section shown in diagrams (a) and (b) above are to be made from a material for which the greatest permissible stresses are +100 N/mm² (tension) and -70 N/mm² (compression).

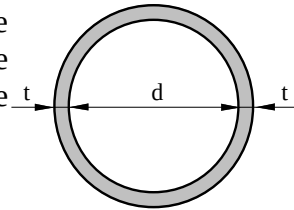
In each case, what are the largest positive and negative bending moments that the section can carry?

Ans : (a) +15.2 kNm, -19.9 kNm
(b) +12.9 kNm, -9.0 kNm

5. A steel pipeline has a circular cross-section, inside diameter (d) 300mm and thickness (t) 8mm, and carries a fluid of relative density 1.04, in addition to its own weight. The pipeline spans a gap over a canal of 12.0m; it may be assumed to run full and to be effectively simply-supported at both ends.

Take the relative density for steel to be 7.80, and note that the second moment of area for a circular tube can be found from the result for a solid circle (see handout) - by deducting the hole in the middle.

$$I = \left\{ \frac{\pi(d + 2t)^4}{64} \right\} - \left\{ \frac{\pi d^4}{64} \right\}$$

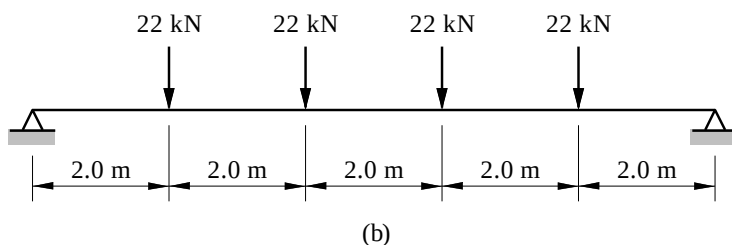
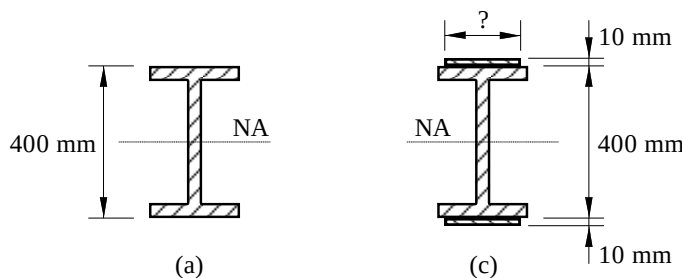


What is greatest bending moment in the span and what is the maximum stress in the steel?

Ans : $M = +23.642 \text{ kNm}$, $s = \pm 40.67 \text{ N/mm}^2$.

6. (a) The symmetrical I-section shown in diagram (a) is 400mm deep and has a second moment of area about its centroidal axis $I = 1.80 \times 10^8 \text{ mm}^4$. If the stress in the section must not exceed $\pm 100 \text{ N/mm}^2$, what is the largest bending moment that it can carry?

(b) The section is to be used for a simply-supported beam, spanning 10.0m and carrying four concentrated loads of 22 kN each, as shown in diagram (b).

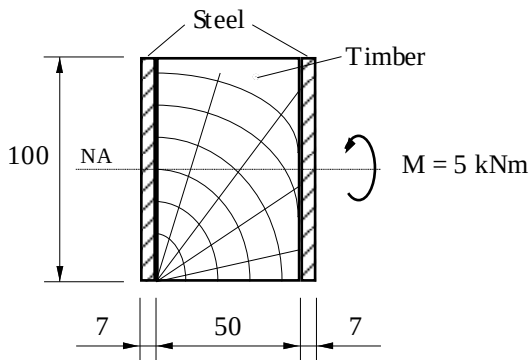


Draw the bending moment diagram for the beam. What is the largest bending moment that must be carried? Define the central region of the span within which the section will require reinforcement in order to carry the load.

(c) The central portion of the span is to be reinforced by welding 10mm plates, made from the same steel as the I-section, to the upper and lower flanges as shown in diagram (c). What is the minimum width of plates required?

Ans : $M = 90 \text{ kNm}$, $M_{\max} = 132 \text{ kNm}$, $x = 2.091 \rightarrow 7.909 \text{ m}$, $b = 115.6 \text{ mm}$.

7. A timber beam, of which the 50 x 100 mm cross-section is shown in the diagram, has been reinforced by glueing steel plates 7 mm thick to the sides of the section. The reinforced beam carries a total bending moment of 5 kNm acting about the neutral axis of the composite section.



It may be assumed that the adhesive is sufficiently strong to ensure that the curvature of the steel plates is identical to the curvature of the timber core. Take Young's Modulus for steel $E_s = 200 \text{ GN/m}^2$ and for timber $E_t = 12 \text{ GN/m}^2$.

What are the proportions of the total bending moment carried by the steel plates (M_s) and the timber core (M_t)? What are the greatest stresses in the steel and in the timber?

Ans : $M_s = 4.118 \text{ kNm}$, $M_t = 0.882 \text{ kNm}$, $s_s = \pm 176.5 \text{ N/mm}^2$, $s_t = \pm 10.6 \text{ N/mm}^2$